

Harshwardhan Saini

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Objective

Software Engineer with experience developing AI agents, backend APIs, serverless cloud applications and distributed systems across AWS & GCP, focused on scalable architectures and cloud-native engineering.

Experience

DT Intern

GE Appliances (Haier) | Jan 2026 – June 2026

- Built AI conversational agents (Google ADK, Dialogflow CX) enabling natural-language querying of manufacturing data across AWS, GCP and SharePoint, reducing reliance on manual dashboards.
- Improved LLM-generated SQL reliability by enriching schema metadata and prompt context, enabling accurate cross-project queries across normalized BigQuery datasets.
- Designed backend APIs, cloud infrastructure and CI/CD pipelines supporting AI agents, MCP integrations and multi-agent orchestration.
- Delivered end-to-end features across Angular, Cloud Run, AWS Lambda, Google Cloud Functions and reusable Angular UI libraries.
- Engineered an MCP-based system enabling natural language access to BigQuery, with user-configurable agent behaviour via OpenWebUI.
- Extended GitHub Actions, Jenkins and Terraform CI/CD workflows to automate deployments and infrastructure provisioning for new services, while standardizing workflows for new repositories.

Projects

National Cyber Threat Intelligence Hub (Team project)

- Developed a microservices-based cyber threat intelligence platform processing real-time honeypot events through an event-driven Kafka pipeline.
- Designed and implemented backend microservices, JWT-based authentication and an API Gateway using Express.js and Django.
- Engineered near real-time ETL pipelines that transformed raw honeypot events into normalized datasets for threat analysis.
- Built an event-driven threat analysis workflow that classified attacks using the previous minute of activity from the same source IP, improving contextual threat detection.
- Integrated backend services with a React dashboard using WebSockets for near real-time threat monitoring and deployed the platform on Minikube with Jenkins-based CI/CD automation.

Sage AI (Personal Project) - [Live](#)

- Built a full-stack AI platform using MERN, FastAPI, Redis, and MongoDB Atlas Vector Search, featuring chat, project-level RAG, semantic retrieval, and a provider-agnostic LLM integration layer.
- Designed project workspaces, reusable prompt libraries, and interactive AI-generated quizzes with contextual feedback tailored to each user's response.
- Implemented JWT & OAuth 2.0 authentication, real-time SSE streaming, and asynchronous background embedding workers, integrating a FastAPI AI service for low-latency LLM inference and semantic search.
- Engineered Redis-backed session management, caching, rate limiting, and an asynchronous embedding pipeline to improve scalability, reduce response latency, and support multi-device authentication.
- Designed a cost-efficient deployment architecture using free-tier cloud services (Vercel, Render, MongoDB Atlas, Upstash Redis) to support scalable AI workloads.

Education

NIIT University | Btech in Computer Science & Engineering | 2022-26

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, Go
- **Frontend:** HTML5, CSS3, Tailwind CSS, React.js, Next.js, Angular
- **Backend Technologies:** Node.js, Express.js, Flask, FastAPI, Django, RESTful APIs
- **Databases (SQL & NoSQL):** MySQL, PostgreSQL, MongoDB, Redis
- **Cloud:** AWS, Google Cloud Platform (GCP)
- **DevOps & Infrastructure:** Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git
- **Agentic AI & Gen AI:** Google ADK, Dialogflow CX, LangChain, LLM APIs, Retrieval-Augmented Generation (RAG), Vector Databases, Model Context Protocol (MCP), Prompt Engineering
- **Systems & Architecture:** Microservices, API Gateway, Kafka, WebSockets, Multi-Agent Systems